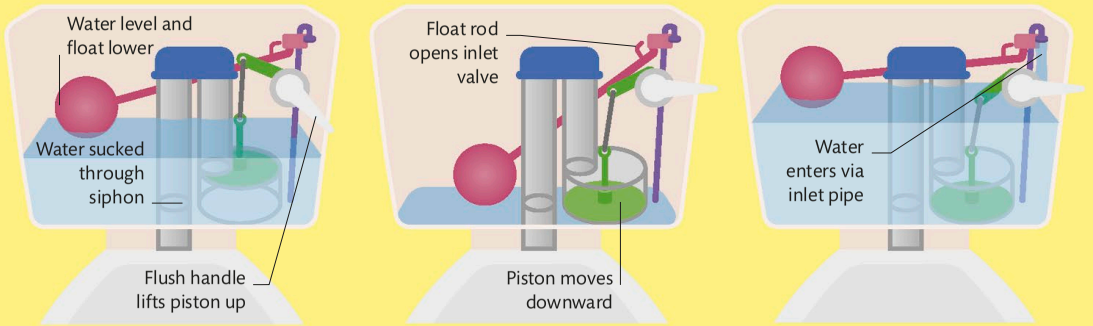




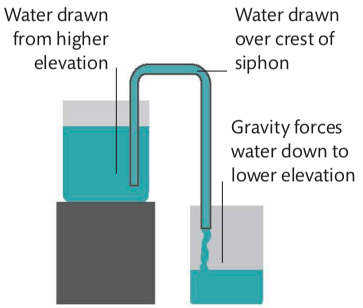
1 Flushing
Turning the handle moves a lever to lift the piston upward. This forces water through the siphon tube, generating a suction force that pulls the rest of the cistern's water through the siphon and out into the bowl.

2 Emptying
The cistern empties rapidly, and water flows around the bowl and out through the drainpipe, carrying away the waste. The piston drops, and the float ball sinks, moving the float rod, which opens the inlet valve.

3 Refilling
With the inlet valve open, water enters the cistern. As the water level rises, so does the float ball. The float moves the float rod to close the inlet valve once the cistern has filled to the required level.

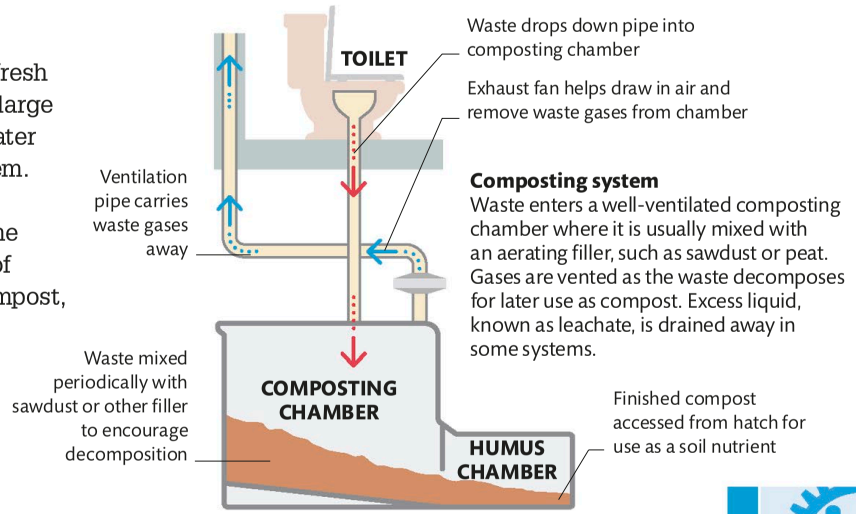
SIPHONS

Many toilets use siphoning to move water from the cistern to the toilet bowl or from the bowl to the drainpipe. Once some water is forced through the highest point of the upturned, U-shaped siphon, gravity and cohesive forces in the liquid help continue the siphoning action until there is no water left.



Composting toilets

At 1⅗–4¾ gallons (6–18 liters) per flush, a standard toilet's fresh water consumption can mount up over time, especially in a large household. In contrast, composting toilets use little or no water and do not place any demands on a municipal sewage system. Instead, these self-contained systems rely on the process of aerobic decomposition, in which bacteria, fungi, and, in some systems, earthworms break down the waste over a period of weeks or months into harmless, largely odor-free humus compost, which can be used as a natural fertilizer.



Composting system
Waste enters a well-ventilated composting chamber where it is usually mixed with an aerating filler, such as sawdust or peat. Gases are vented as the waste decomposes for later use as compost. Excess liquid, known as leachate, is drained away in some systems.

2.3 BILLION PEOPLE DO NOT HAVE BASIC SANITATION FACILITIES